

Laplace Transformation

5. Second Shifting Theorem:

$$L\{f(t-a)u(t-a)\} = e^{-as}F(s)$$

$$f(t-a)u(t-a) = L^{-1}\{e^{-as}F(s)\}$$

$$L\{f(t)u(t-a)\} = e^{-as}L\{f(t+a)\}$$

$$f(t)u(t-a) = L^{-1}\{e^{-as}L\{f(t+a)\}\}$$

Question: $f(t) = t^2$ when $1 < t < 2$

Solution: $f(t) = t^2u(t-1) - t^2u(t-2)$

$$L\{f(t)\} = L\{t^2u(t-1)\} - L\{t^2u(t-2)\}$$

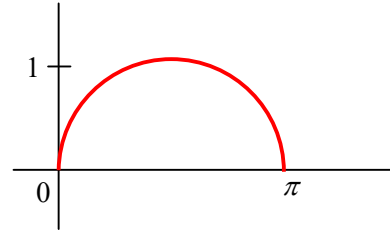
$$\left\{ \because L\{t^2\} = \frac{2!}{s^3} \right\}, \quad L\{f(t)\} = e^{-s}L\{(t+1)^2\} - e^{-2s}L\{(t+2)^2\}$$

Question: $f(t) = \sin t \{u(t-0) - u(t-\pi)\}$

Solution: $L\{f(t)\} = e^0 L\{\sin(t)\} - e^{-\pi s} L\{\sin t + \pi\}$

$$L\{f(t)\} = \frac{1}{s^2 + 1} + e^{-\pi s} L\{\sin t\}$$

$$L\{f(t)\} = \frac{1}{s^2 + 1} + \frac{e^{-\pi s}}{s^2 + 1} = \frac{1 + e^{-\pi s}}{(s^2 + 1)}$$



Impulse function:

$$L\{\delta(t-a)\} \quad \text{for } t > a, \quad \delta(t-a) = \infty, \quad t = a$$

$$= \int_0^{\infty} e^{-st} \delta(t-a) dt$$

$$L\{\delta(t-a)\} = e^{-sa}$$

Example of A forced oscillator

Question: $y'' + 3y' + 2y = r(t)$

$$\text{And } r(t) = u(t-1) - u(t-2)$$

$$y(0) = 0, \quad y'(0) = 0$$

Solution: $L\{y'' + 3y' + 2y\} = L\{u(t-1) - u(t-2)\}$

$$s^2 L(y) - sy(0) - f'(0) + 3s(L(y)) - f'(0) + 2L\{y\} = L\{u(t-1)\} - L\{u(t-2)\}$$

$$s^2 y + 3sy + 2y = \frac{e^{-s}}{s} - \frac{e^{-2s}}{s}$$

$$y = \frac{1}{(s^2 + 3s + 2)} \left(\frac{e^{-s}}{s} - \frac{e^{-2s}}{s} \right)$$

Hammer blow Response of a mass-spring system:

Question: $y'' + 3y' + 2y = \delta(t-1)$, $y(0) = 0$ & $y'(0) = 0$

Solution: $L\{y'' + 3y' + 2y\} = L\{\delta(t-1)\}$

$$= L\{y\}(s^2 + 3s + 2) = e^{-s}$$

$$L\{y\} = \frac{e^{-s}}{(s^2 + 3s + 2)}$$

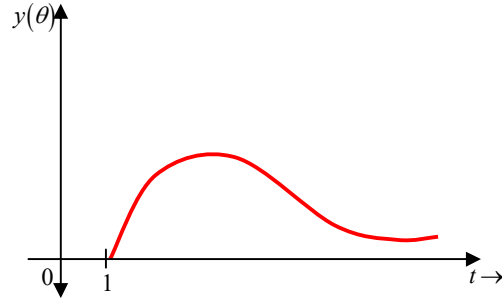
$$L\{y\} = \left(\frac{1}{(s+1)(s+2)} \right) e^{-s} = \left[\frac{1}{(s+1)} - \frac{1}{(s+2)} \right] e^{-s}$$

$$L^{-1}\{L(y)\} = L^{-1}\left\{ \frac{e^{-s}}{(s+1)} - \frac{e^{-s}}{(s+2)} \right\}$$

$$L^{-1}\{L(y)\} = e^{-(t-1)}u(t-1) - e^{-2(t-1)}u(t-1)$$

By using the second shift theorem

$$y(t) = \begin{cases} 0 & 0 < t < 1 \\ e^{-(t-1)} - e^{-2(t-1)} & t > 1 \end{cases}$$



Question: Find the output voltage response when $R = 20\Omega$, $L = 1H$ & $C = 10^{-4}F$.

The LCR circuit is a forced oscillator. $q(0) = i(0) = 0$

Solution: $L \frac{di}{dt} + iR + \frac{q}{C} = \delta(t)$

$$\Rightarrow L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = \delta(t)$$

Now take $L \Rightarrow \frac{d^2q}{dt^2} + \frac{R}{C} \frac{dq}{dt} + \frac{q}{LC} = \frac{\delta(t)}{L}$ aplace both side

$$\Rightarrow L \left\{ \frac{d^2q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{q}{LC} \right\} = \frac{\delta(t)}{L}$$

$$\Rightarrow s^2 L \{q\} - \delta q(0) - q'(0) + \frac{R}{C} s L \{q\} - \frac{R}{L} q(0) + \frac{1}{LC} L \{q\} = \frac{1}{L} L \{\delta(t)\}$$

$$\Rightarrow s^2 Q + \frac{RSQ}{L} + \frac{1}{LC} Q = \frac{1}{L} e^{-0}$$

$$\Rightarrow Q = \frac{1/L}{\left(s^2 + \frac{RS}{L} + \frac{1}{LC} \right)} = \frac{LC}{(LCS^2 + RCS + 1)}$$

$$S = \frac{-RC \pm \sqrt{R^2 C^2 - 4LC}}{2LC}$$

$$Q = \frac{LC}{\left(\frac{S - RC + \sqrt{(RC)^2 - 4LC}}{2LC} \right) \left(\frac{S - RC - \sqrt{(RC)^2 - 4LC}}{2LC} \right)}$$

$$Q = \frac{2L^2C^2}{2\sqrt{(RC)^2 - 4LC}} \left[\frac{1}{S - RC - \sqrt{(RC)^2 - 4LC}} - \frac{1}{S - RC + \sqrt{(RC)^2 - 4LC}} \right]$$

$$L^{-1} = q = \frac{2L^2C^2}{2\sqrt{(RC)^2 - 4LC}} \cdot \left[e^{\left(RC + \sqrt{(RC)^2 - 4LC} \right)t} - e^{t \left(RC - \sqrt{(RC)^2 - 4LC} \right)} \right]$$

$$L = 1H$$

$$R = 20\Omega \quad C = 10^{-4}F$$

$$q = \frac{2 \times 10^{-8}}{2\sqrt{(20 \times 10^{-4})^2 - 4 \times 10^{-4}}} \left[e^{Rct} \left(e^{\left(\sqrt{(RC)^2 - 4LC} \right)t} - e^{-\sqrt{(RC)^2 - 4LC}} \right) \right]$$

$$q = \frac{10^{-8}}{\sqrt{20 \times 20 \times 10^{-8} - 4 \times 10^{-4}}} e^{20 \times 10^{-4}t} = \frac{10^{-8}}{\sqrt{20 \times 20 \times 10^{-8} - 4 \times 10^{-4}}} e^{20 \times 10^{-4}t}$$

Important: $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$